# Supplementary Material for apcMIA: Adaptive Perturbation-assisted Contrastive Membership Inference Attacks

## Outline of the Supplementary Material

This supplementary material provides additional methodological details, experimental settings, and extended results supporting the main manuscript. The organization is as follows.

| Section | Description |
|---|---|
| A | Provides the training algorithm for `apcMIA`. |
| B | Describes the evaluation setup, including datasets, dataset splits, target-model architectures, training configurations, evaluation metrics, and attack baselines. |
| C | Reports additional attack-performance results for image architectures not fully shown in the main paper, including AdvCNN and VanCNN. |
| D | Provides additional ROC results across datasets and model architectures to complement the main low-FPR evaluation. |
| E | Presents bar-plot comparisons of TPR at low-FPR regimes across target model families. |
| F | Reports membership advantage results at 0.1% and 1.0% FPR across datasets and architectures. |
| G | Analyzes prediction-uncertainty and entropy distributions before and after `apcMIA` perturbation. |
| H | Evaluates the robustness of `apcMIA` under DP–SGD-trained target models. |
| I | Describes the gap-aware sigmoid schedule used to adapt learning behavior based on target-model generalization. |
| J | Shows the evolution of learned thresholds and attack loss during training across image and non-image datasets. |
| K | Provides t-SNE visualizations showing how perturbation changes the separability of member and non-member samples in the latent space. |

# A Algorithm

**Algorithm S1** `apcMIA` Training Loop: Jointly optimize $\mathcal{F}_{\text{attack}}$, $\mathcal{F}_{\text{perturb}}$, and thresholds $\tau_c, \tau_e$ using member/non-member shadow data.

**Require:** Shadow outputs $(\mathbf{p}_i^m, \mathbf{p}_i^{nm}, y_i)$, model parameters $\theta_a, \theta_p$, thresholds $\tau_c, \tau_e$, learning rates $\eta_a, \eta_p, \eta_c, \eta_e$
**Ensure:** Trained attack model $\mathcal{F}_{\text{attack}}$

**for** each batch from $\mathcal{D}^s$ **do**
  **Cosine Similarity Thresholding**
  $s_i \leftarrow \max_j \cos(\mathbf{p}_i^{nm}, \mathbf{p}_j^m)$
  $z_i^c \leftarrow \text{GumbelSoftmax}\left([-(s_i - \sigma(\tau_c))k_c,\ (s_i - \sigma(\tau_c))k_c]\right)$
  **Entropy Thresholding**
  $H(\mathbf{p}_i^{nm}) \leftarrow -\sum_{c=1}^{C} \mathbf{p}_{i,c}^{nm} \log(\mathbf{p}_{i,c}^{nm} + \epsilon)$
  $q_e \leftarrow \text{Quantile}(\{H(\mathbf{p}_i^{nm}) \mid z_i^c = 1\},\ \sigma(\tau_e))$
  $z_i^e \leftarrow \sigma(H(\mathbf{p}_i^{nm}) - q_e)$
  **Mask and Perturbation**
  $z_i \leftarrow z_i^c \cdot z_i^e$
  $\Delta\mathbf{p}_i^{nm} \leftarrow \mathcal{F}_{\text{perturb}}(\mathbf{p}_i^{nm}; \theta_p)$
  $\hat{\mathbf{p}}_i^{nm} \leftarrow \text{clip}(\mathbf{p}_i^{nm} + \alpha \cdot z_i \cdot \Delta\mathbf{p}_i^{nm},\ \epsilon,\ 1)$
  Normalize $\hat{\mathbf{p}}_i^{nm} \leftarrow \hat{\mathbf{p}}_i^{nm} / \sum_c \hat{\mathbf{p}}_{i,c}^{nm}$
  **Batch Construction**
  $\mathbf{p}'_i \leftarrow \begin{cases} \hat{\mathbf{p}}_i^{nm}, & \text{if } y_i = 0 \\ \mathbf{p}_i^m, & \text{if } y_i = 1 \end{cases}$
  **Attack Loss**
  $\hat{y}_i \leftarrow \mathcal{F}_{\text{attack}}(\mathbf{p}'_i; \theta_a)$
  $\mathcal{L}_{\text{attack}} \leftarrow -\frac{1}{N} \sum_i [y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$
  **Contrastive Loss ($\mathbf{CL_p}$)**
  $\mathbf{e}_i \leftarrow \text{PenultimateLayer}(\mathcal{F}_{\text{attack}}(\mathbf{p}'_i))$
  $\mathcal{L}_{\text{CL}_p} \leftarrow \text{ContrastiveLoss}(\{\mathbf{e}_i\}, \{y_i\})$
  **Total Loss and Updates**
  $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{attack}} + \lambda \cdot \mathcal{L}_{\text{CL}_p}$
  $\theta_a \leftarrow \theta_a - \eta_a \cdot \nabla_{\theta_a} \mathcal{L}_{\text{total}}$
  $\theta_p \leftarrow \theta_p - \eta_p \cdot \nabla_{\theta_p} \mathcal{L}_{\text{total}}$
  $\tau_c \leftarrow \tau_c - \eta_c \cdot \nabla_{\tau_c} \mathcal{L}_{\text{total}}$
  $\tau_e \leftarrow \tau_e - \eta_e \cdot \nabla_{\tau_e} \mathcal{L}_{\text{total}}$
**end for**
**return** $\mathcal{F}_{\text{attack}}$

# B Evaluation

## B.1 Datasets

We evaluate `apcMIA` across nine widely used benchmark datasets spanning both image and tabular domains. This diversity allows us to assess the generality and robustness of our attack under various data modalities and model families. For fair comparison, all image datasets are resized to $64 \times 64$ resolution.

- **UTKFace [1].** UTKFace is a face dataset containing over 23,000 images annotated with age, gender, and race. We focus on four racial groups: White, Black, Asian, and Indian, resulting in 22,012 samples. For each of the target and shadow models, we use 5,500 member samples for training and 5,500 held-out non-member samples, yielding 11,000 PVs per model. The member and non-member subsets used for the target and shadow models are mutually exclusive.
- **CIFAR [2].** CIFAR-10 and CIFAR-100 each contain 60,000 color images of resolution $32 \times 32$, distributed across 10 and 100 classes, respectively. Each dataset consists of 50,000 training and 10,000 test images. For each dataset, the target and shadow models are each trained using 14,770 member samples and evaluated using 14,770 held-out non-member samples, yielding 29,540 PVs per model. The four member and non-member subsets used for the target and shadow models are sampled without overlap.
- **STL-10 [3].** STL-10 comprises 1,300 images per class across 10 categories. It offers higher-resolution and more varied content than CIFAR but contains fewer samples per class, making it a relatively challenging dataset. The target and shadow models are each trained using 3,250 member samples and evaluated using 3,250 held-out non-member samples, yielding 6,500 PVs per model. The four subsets are mutually exclusive.
- **Fashion-MNIST [4].** Fashion-MNIST contains 70,000 grayscale images of resolution $28 \times 28$, distributed across 10 fashion categories. The target and shadow models are each trained using 15,000 member samples and evaluated

using 15,000 held-out non-member samples, yielding 30,000 PVs per model. The member and non-member subsets used for the two models are sampled without overlap.

To evaluate the robustness of `apcMIA` beyond vision tasks, we include four widely used non-image datasets spanning geolocation, purchasing behaviour, healthcare, and demographic records:

- **Texas-100.** This dataset is based on inpatient discharge data collected from Texas hospitals between 2006 and 2009.[1] It contains attributes such as diagnoses, procedures, external causes of injury, and demographic variables, including age, gender, race, hospital ID, and length of stay. We use the processed 100-class variant from [5], which contains 67,330 records represented by 6,170 binary features. The target and shadow models are each trained using 5,000 member samples and evaluated using 5,000 held-out non-member samples, yielding 10,000 PVs per model.
- **Location [6].** We use the processed version of the Foursquare check-in dataset.[2] Following [5], users with fewer than 25 check-ins and venues with fewer than 100 visits are filtered out, yielding 5,010 user profiles represented by 446 binary features. The records are clustered into 30 geosocial classes. The target and shadow models are each trained using 1,252 member samples and evaluated using 1,252 held-out non-member samples, yielding 2,504 PVs per model. The four subsets are mutually exclusive.
- **Purchase-100.** This dataset is a processed version of the "Acquire Valued Shoppers" challenge dataset. It contains 197,324 customer records, each represented by 600 binary features. Following [5], the samples are grouped into 100 classes, with each class representing a distinct purchasing pattern. The target and shadow models are each trained using 20,000 member samples and evaluated using 20,000 held-out non-member samples, yielding 40,000 PVs per model.
- **Adult (Census Income).** The Adult dataset contains 48,842 records with 14 demographic and employment-related attributes, including age, education, marital status, and occupation.[3] The binary classification task predicts whether an individual's annual income exceeds $50K. The target and shadow models are each trained using 6,105 member samples and evaluated using 6,105 held-out non-member samples, yielding 12,210 PVs per model.

For all datasets, member samples are used to train the corresponding target or shadow model, whereas non-member samples are held out from that model's training. The resulting member and non-member PVs are balanced within each model partition. The shadow-model PVs are used to train the attack, while the target-model PVs are used to evaluate it. Table S1 summarizes the dataset partitions and training configurations. All target and shadow models are trained for up to 60 epochs, with early stopping based on validation loss.

TABLE S1
Summary of dataset partitions and target/shadow-model training configurations. The member and non-member columns report mutually exclusive subsets. The "Total Used" column is the sum of the four target and shadow partitions.

| Dataset | # Classes | Resolution | Max. Epochs | Target Model | | Shadow Model | | Total Used |
|---|---|---|---|---|---|---|---|---|
| | | | | Member | Non-member | Member | Non-member | |
| UTKFace | 4 | $64 \times 64$ | 60 | 5,500 | 5,500 | 5,500 | 5,500 | 22,000 |
| Fashion-MNIST | 10 | $64 \times 64$ | 60 | 15,000 | 15,000 | 15,000 | 15,000 | 60,000 |
| STL-10 | 10 | $64 \times 64$ | 60 | 3,250 | 3,250 | 3,250 | 3,250 | 13,000 |
| CIFAR-10 | 10 | $64 \times 64$ | 60 | 14,770 | 14,770 | 14,770 | 14,770 | 59,080 |
| CIFAR-100 | 100 | $64 \times 64$ | 60 | 14,770 | 14,770 | 14,770 | 14,770 | 59,080 |
| Location | 30 | N/A | 60 | 1,252 | 1,252 | 1,252 | 1,252 | 5,008 |
| Purchase-100 | 100 | N/A | 60 | 20,000 | 20,000 | 20,000 | 20,000 | 80,000 |
| Texas-100 | 100 | N/A | 60 | 5,000 | 5,000 | 5,000 | 5,000 | 20,000 |
| Adult | 2 | N/A | 60 | 6,105 | 6,105 | 6,105 | 6,105 | 24,420 |

TABLE S2
Training and Testing Accuracy of Target Models Across Datasets and Architectures. Values are reported as *train/test*. Image datasets use WideResNet, AdvCNN, and VanCNN; non-image datasets use MLP. Models are categorized into generalization regimes based on the train–test accuracy gap $g = |T_{tr} - T_{te}|$: **G** ($g \leq 0.11$), **M** ($0.11 < g \leq 0.18$), and **O** ($g > 0.18$).

| Target Model | FMNIST | UTKFace | STL-10 | CIFAR-10 | CIFAR-100 |
|---|---|---|---|---|---|
| WideResNet | 0.9654/0.9166 **G** (0.0488) | 0.9516/0.8547 **G** (0.0970) | 0.9855/0.8262 **M** (0.1593) | 0.9706/0.9020 **G** (0.0686) | 0.9254/0.6507 **O** (0.2747) |
| AdvCNN | 0.9564/0.9274 **G** (0.0290) | 0.9869/0.8647 **M** (0.1222) | 0.9455/0.7308 **O** (0.2147) | 0.9753/0.8827 **G** (0.0926) | 0.9497/0.5190 **O** (0.4307) |
| VanCNN | 0.9843/0.9100 **G** (0.0743) | 0.9892/0.8520 **M** (0.1372) | 0.8306/0.5985 **O** (0.2321) | 1.0000/0.6953 **O** (0.3047) | 0.6731/0.3027 **O** (0.3704) |

| Target Model | Location | Purchase-100 | Texas-100 | Adult |
|---|---|---|---|---|
| MLP | 0.7786/0.6335 **M** (0.1452) | 0.9621/0.8593 **G** (0.1028) | 0.6277/0.5708 **G** (0.0569) | 0.8507/0.8477 **G** (0.0030) |

1. https://www.dshs.texas.gov/THCIC/Hospitals/Download.shtm
2. https://sites.google.com/site/yangdingqi/home/foursquare-dataset
3. http://archive.ics.uci.edu/ml/datasets/Adult

### B.1.1 Target Models

We train shadow and target models separately for each dataset. On image datasets, we evaluate three architectures: VanCNN, a custom lightweight 3-layer CNN [7]; AdvCNN adopted from [8], a deeper VGG-style model with 9 convolutional layers (64 $\rightarrow$ 512 channels) and dropout before the 8192–4096–output fully connected head; and WideResNet variant with four residual blocks per stage and widths of 32, 64, and 128 adopted from [9]. All CNNs are optimized using SGD with momentum 0.9 and weight decay $5 \times 10^{-4}$, and are trained for up to 60 epochs with early stopping based on validation loss. Purchase-100 and Texas-100 employ deep 6-layer networks with layer normalization and dropout (rate 0.5), while Adult uses a 3-layer MLP and Location a compact 2-layer design. All MLPs follow the same maximum training budget of 60 epochs with early stopping. Purchase and Texas are trained with a learning rate of $1 \times 10^{-2}$, multi-step decay and weight decay $1 \times 10^{-4}$; Adult uses a lower rate of $1 \times 10^{-3}$ with the same decay and scheduler.

Following the training of these target models, the resulting train and test accuracies are reported in Table S2, showing the generalization gap across datasets and architectures. We chose these configurations to produce a range of target model complexities and train-test accuracy gaps, allowing us to evaluate the effectiveness of `apcMIA` across models with varying levels of generalization.

In addition to the target models, `apcMIA` employs two MLPs: 1) The perturbation model ($\mathcal{F}_p$) which is a 5-layer MLP that generates adversarial perturbations to modify prediction vectors, 2) The attack model ($\mathcal{F}_a$) which is a 4-layer MLP that takes perturbed and non-perturbed PVs as input and outputs membership predictions. Both models are trained using Adam optimizer with a learning rate of $1 \times 10^{-3}$. The two learnable thresholds, the cosine similarity threshold $\tau_c$ and the entropy quantile threshold $\tau_e$, are initialized to 0.5 and jointly optimized using separate Adam optimizers with a learning rate of $1 \times 10^{-2}$.

### B.1.2 Metrics

In this paper, we evaluate our method using the following metrics:

- **Full Log-scale ROC.** Receiver Operating Characteristics (ROC) are commonly used curves that show the trade-off between true positive rate (TPR) and false positive rate (FPR). However, in sensitive privacy settings, Low FPR matters the most [10]. Therefore, we compute the logarithmic scale of ROC to emphasise low FPR regions.
- **TPR at Low FPR.** We report the TPR at FPR values of $0.1\%$ and $1.0\%$. TPR@$\beta$ FPR is obtained by linear interpolation of the empirical ROC curve at $\beta$, where $\beta \in \{0.001, 0.01\}$ [10].
- **Membership Advantage (MA).** We additionally report membership advantage [11], defined as the vertical distance between the attack ROC curve and the random-guess baseline. MA captures how much better the attack performs than chance and is particularly informative in low-FPR regions where small improvements can imply significant privacy risk.
- **Balanced Accuracy and AUC.** These are two standard average-case metrics commonly used to evaluate binary classification tasks. Most MIAs [12]–[17] are binary classifiers, so these metrics are often reported. Although these average-case metrics can obscure behavior in low-FPR regions, we report them for completeness and comparison with prior work.

### B.1.3 Attack Baselines

We compare `apcMIA` against five established membership inference baselines that cover several attack styles. These include NSH [18], which operates directly on posterior vectors; meMIA [13], which leverages embeddings from expert MLPs; seqMIA [15], which models temporal correlations across predictions; LiRA [10][4], a likelihood-ratio attack designed for strong low-FPR performance; and RMIA [9][5], which further strengthens this line through a statistically grounded and computationally efficient likelihood-ratio test.

Our single-shadow design is specific to `apcMIA` and is not imposed on LiRA or RMIA. For these two baselines, we follow their original evaluation protocols rather than constraining them to the computational budget of `apcMIA`. Across all nine datasets and all evaluated dataset–architecture configurations, LiRA is evaluated in the online setting using 256 reference models under an IN/OUT construction. We use stable logit scaling of the confidence-based statistic, Gaussian likelihood-ratio evaluation, and the 18-query multi-query setting. RMIA is evaluated using its standard online multi-model protocol with 254 reference models, comprising 127 IN and 127 OUT models, together with the same 18-query setting and the recommended parameterization ($\gamma = 2$). For the remaining baselines, we follow the settings described in their original papers and use the corresponding published implementations where available. In comparison, `apcMIA` trains a single shadow model together with its perturbation and attack networks. At inference, `apcMIA` requires only one target-model evaluation returning the complete posterior vector for each candidate; the shadow model is not queried during inference.

4. https://github.com/tensorflow/privacy/tree/master/research/mi_lira_2021
5. https://github.com/privacytrustlab/ml_privacy_meter/tree/d32734161a3395211fe5f3cd461932290b1fafbe/research/2024_rmia

# C Additional Attack Results for Other Architectures

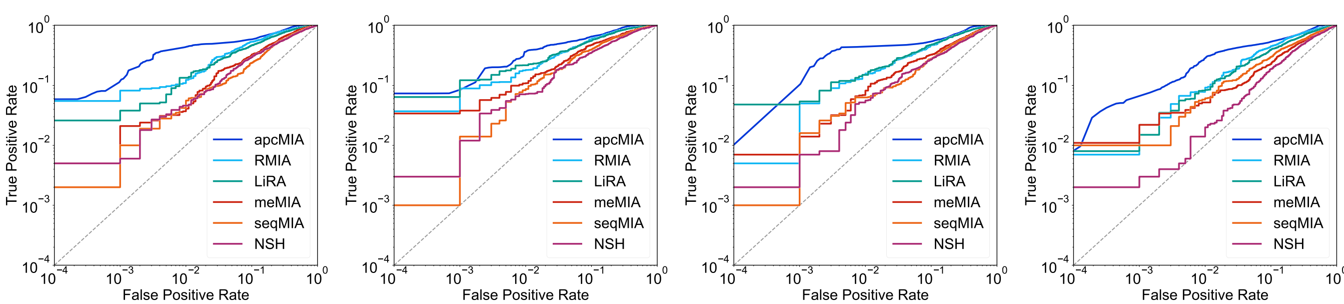
TABLE S3
Attack comparison for **advCNN**. Results are mean ± std over 5 independent runs.

| Method | CIFAR-10 | STL-10 | CIFAR-100 | UTKFace | FMNIST |
|---|---|---|---|---|---|
| **Metric: TPR @ 0.1% FPR** | | | | | |
| RMIA [9] | 6.7 ± 0.52% | 8.1 ± 0.61% | 1.2 ± 0.19% | 4.9 ± 0.43% | 1.4 ± 0.22% |
| LiRA [10] | 3.1 ± 0.38% | 8.1 ± 0.58% | 1.2 ± 0.17% | 4.9 ± 0.46% | 1.1 ± 0.18% |
| meMIA [13] | 1.2 ± 0.25% | 3.8 ± 0.42% | 0.5 ± 0.10% | 1.4 ± 0.26% | 1.1 ± 0.16% |
| seqMIA [15] | 0.6 ± 0.14% | 1.3 ± 0.20% | 0.5 ± 0.12% | 1.4 ± 0.23% | 0.8 ± 0.13% |
| NSH [18] | 0.4 ± 0.09% | 1.1 ± 0.15% | 0.1 ± 0.06% | 0.7 ± 0.11% | 0.2 ± 0.08% |
| **apcMIA** | **11.8** ± 0.45% | **8.4** ± 0.55% | **1.3** ± 0.16% | **10.3** ± 0.48% | **6.5** ± 0.39% |
| **Metric: Balanced Accuracy** | | | | | |
| RMIA [9] | 0.760 ± 0.010 | 0.783 ± 0.012 | 0.667 ± 0.015 | 0.733 ± 0.011 | 0.723 ± 0.014 |
| LiRA [10] | 0.730 ± 0.012 | 0.776 ± 0.013 | 0.649 ± 0.016 | 0.729 ± 0.013 | 0.691 ± 0.015 |
| meMIA [13] | 0.672 ± 0.014 | 0.718 ± 0.015 | 0.636 ± 0.017 | 0.661 ± 0.014 | 0.623 ± 0.016 |
| seqMIA [15] | 0.650 ± 0.015 | 0.703 ± 0.016 | 0.626 ± 0.018 | 0.650 ± 0.015 | 0.623 ± 0.017 |
| NSH [18] | 0.636 ± 0.016 | 0.675 ± 0.017 | 0.621 ± 0.019 | 0.629 ± 0.016 | 0.602 ± 0.018 |
| **apcMIA** | **0.769** ± 0.006 | **0.814** ± 0.008 | **0.737** ± 0.007 | **0.769** ± 0.009 | **0.730** ± 0.005 |
| **Metric: AUC** | | | | | |
| RMIA [9] | 0.829 ± 0.009 | 0.859 ± 0.011 | 0.729 ± 0.014 | 0.804 ± 0.010 | 0.794 ± 0.013 |
| LiRA [10] | 0.805 ± 0.010 | 0.849 ± 0.012 | 0.696 ± 0.015 | 0.802 ± 0.011 | 0.754 ± 0.014 |
| meMIA [13] | 0.732 ± 0.011 | 0.786 ± 0.013 | 0.678 ± 0.016 | 0.721 ± 0.012 | 0.669 ± 0.015 |
| seqMIA [15] | 0.689 ± 0.012 | 0.766 ± 0.014 | 0.667 ± 0.017 | 0.695 ± 0.013 | 0.667 ± 0.016 |
| NSH [18] | 0.678 ± 0.013 | 0.733 ± 0.015 | 0.656 ± 0.018 | 0.676 ± 0.014 | 0.623 ± 0.017 |
| **apcMIA** | **0.879** ± 0.005 | **0.905** ± 0.007 | **0.786** ± 0.006 | **0.863** ± 0.008 | **0.843** ± 0.004 |

TABLE S4
Attack comparison for **VanCNN**. Results are mean ± std over 5 independent runs.

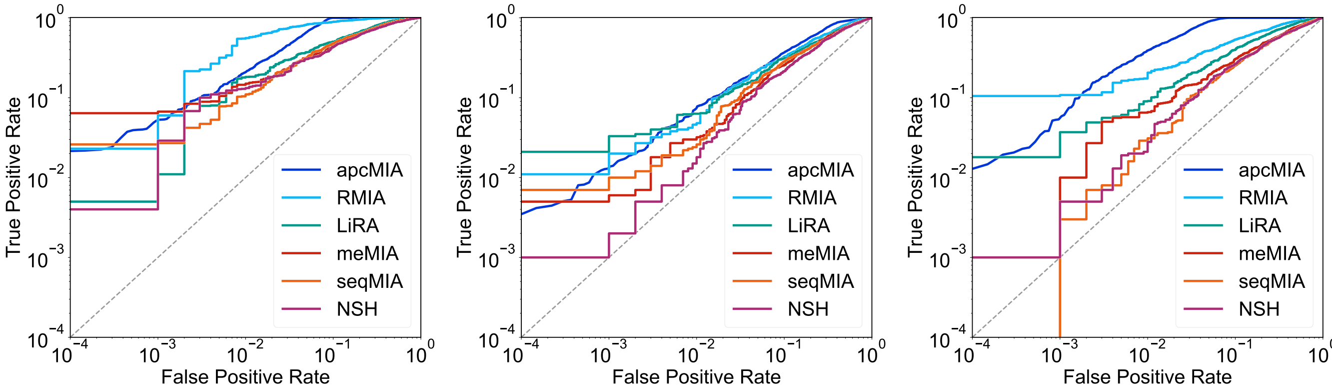
| Method | CIFAR-10 | STL-10 | CIFAR-100 | UTKFace | FMNIST |
|---|---|---|---|---|---|
| **Metric: TPR @ 0.1% FPR** | | | | | |
| RMIA [9] | 7.6 ± 0.58% | 6.8 ± 0.49% | 5.6 ± 0.42% | 1.2 ± 0.20% | 0.6 ± 0.13% |
| LiRA [10] | 4.0 ± 0.41% | 2.9 ± 0.32% | 2.6 ± 0.29% | 0.6 ± 0.12% | 0.6 ± 0.11% |
| meMIA [13] | 4.0 ± 0.37% | 2.8 ± 0.30% | 0.4 ± 0.09% | 0.6 ± 0.14% | 0.4 ± 0.08% |
| seqMIA [15] | 2.2 ± 0.26% | 0.7 ± 0.13% | 0.1 ± 0.05% | 0.6 ± 0.10% | 0.4 ± 0.07% |
| NSH [18] | 1.9 ± 0.22% | 0.7 ± 0.11% | 0.1 ± 0.04% | 0.2 ± 0.06% | 0.1 ± 0.03% |
| **apcMIA** | **37.9** ± 1.12% | **8.8** ± 0.53% | **5.6** ± 0.38% | **1.2** ± 0.15% | **0.9** ± 0.10% |
| **Metric: Balanced Accuracy** | | | | | |
| RMIA [9] | 0.861 ± 0.009 | 0.780 ± 0.011 | 0.746 ± 0.013 | 0.662 ± 0.014 | 0.672 ± 0.015 |
| LiRA [10] | 0.748 ± 0.011 | 0.745 ± 0.012 | 0.703 ± 0.014 | 0.652 ± 0.015 | 0.656 ± 0.016 |
| meMIA [13] | 0.730 ± 0.013 | 0.705 ± 0.014 | 0.646 ± 0.016 | 0.611 ± 0.017 | 0.608 ± 0.018 |
| seqMIA [15] | 0.697 ± 0.014 | 0.653 ± 0.015 | 0.632 ± 0.017 | 0.597 ± 0.018 | 0.597 ± 0.019 |
| NSH [18] | 0.697 ± 0.015 | 0.653 ± 0.016 | 0.628 ± 0.018 | 0.567 ± 0.019 | 0.546 ± 0.020 |
| **apcMIA** | **0.880** ± 0.005 | **0.826** ± 0.007 | **0.959** ± 0.004 | **0.785** ± 0.008 | **0.707** ± 0.006 |
| **Metric: AUC** | | | | | |
| RMIA [9] | 0.937 ± 0.008 | 0.858 ± 0.010 | 0.827 ± 0.012 | 0.717 ± 0.013 | 0.729 ± 0.014 |
| LiRA [10] | 0.816 ± 0.009 | 0.821 ± 0.011 | 0.771 ± 0.013 | 0.699 ± 0.014 | 0.708 ± 0.015 |
| meMIA [13] | 0.795 ± 0.010 | 0.763 ± 0.012 | 0.697 ± 0.015 | 0.654 ± 0.016 | 0.642 ± 0.017 |
| seqMIA [15] | 0.766 ± 0.011 | 0.705 ± 0.013 | 0.673 ± 0.016 | 0.626 ± 0.017 | 0.625 ± 0.018 |
| NSH [18] | 0.764 ± 0.012 | 0.702 ± 0.014 | 0.663 ± 0.017 | 0.585 ± 0.018 | 0.562 ± 0.019 |
| **apcMIA** | **0.950** ± 0.004 | **0.913** ± 0.006 | **0.978** ± 0.003 | **0.860** ± 0.007 | **0.794** ± 0.005 |

# D Additional ROC Results



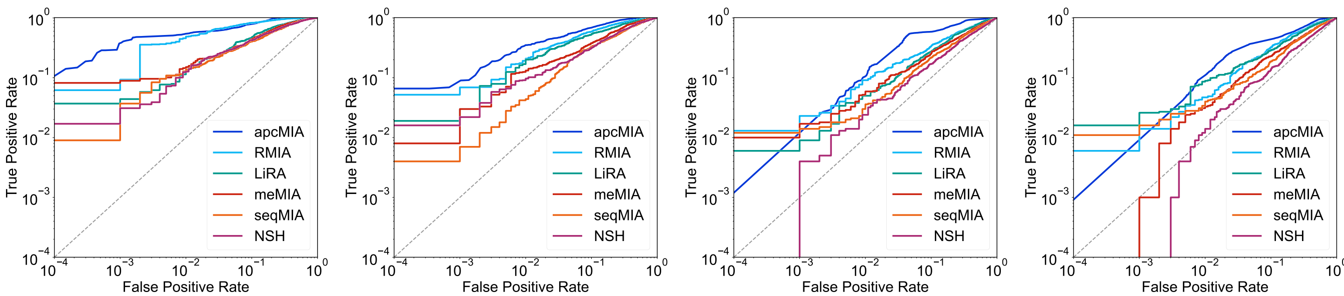

(a) CIFAR-10 (b) STL-10 (c) UTKFace (d) FMNIST

Fig. S1. ROC curves for attacks on four image datasets with the **AdvCNN** backbone.




(a) WideResNet (b) AdvCNN (c) VanCNN

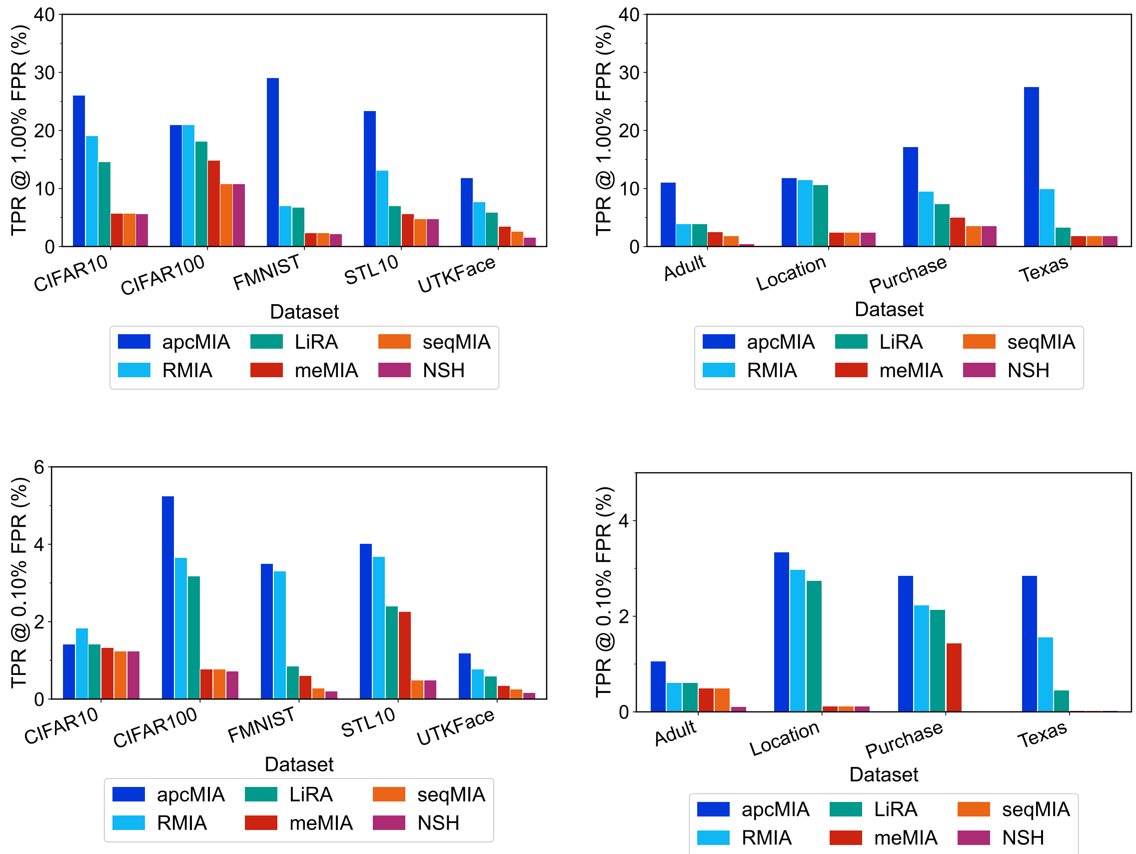
Fig. S2. CIFAR-100 ROC curves across **WideResNet**, **AdvCNN**, and **VanCNN**.




(a) CIFAR-10 (b) STL-10 (c) UTKFace (d) FMNIST

Fig. S3. ROC curves for attacks on four image datasets with the **VanCNN** backbone.

# E Bar Plots for various Attack configurations




(a) WideResNet

(b) MLP

Fig. S4. True Positive Rate (TPR) at 1.0% FPR (top) and 0.1% FPR (bottom) for nine datasets across two target model families: (a) WideResNet and (b) MLP. Overall, `apcMIA` achieves better TPR in both high and low-FPR regimes, demonstrating robustness across image and tabular domains.

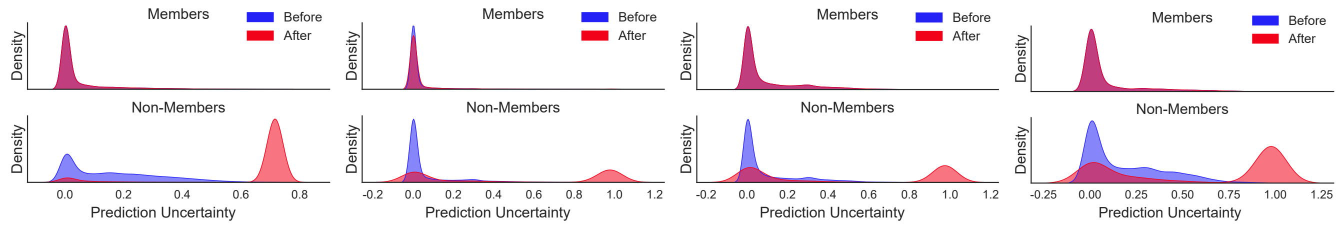
# F Membership Advantage

TABLE S5

Membership Advantage (MA) at 0.1% and 1.0% FPR for nine datasets across four model architectures: WideResNet (WRN), advCNN, vanillaCNN (vanCNN), and MLP. `apcMIA` achieves the highest MA in the majority of settings, with bolded values indicating the best method per row and FPR regime.

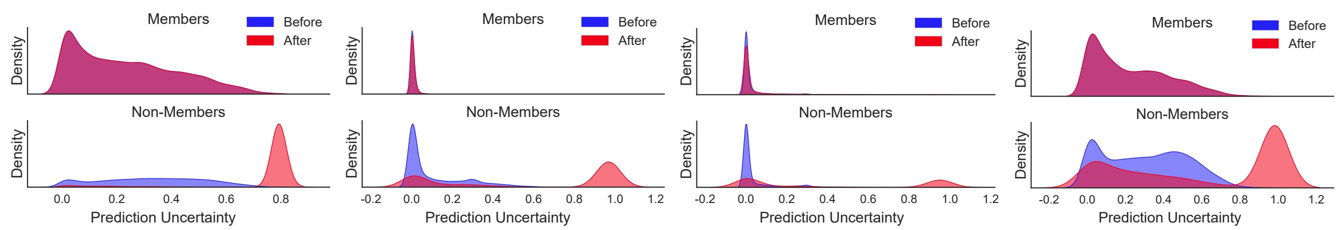
| Dataset | Arch | MA @0.1% FPR | | | | | | MA @1.0% FPR | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | LiRA | NSH | RMIA | meMIA | seqMIA | **apcMIA** | LiRA | NSH | RMIA | meMIA | seqMIA | **apcMIA** |
| CIFAR-10 | vanCNN | 3.9% | 1.8% | 7.5% | 3.9% | 2.1% | **37.8%** | 11.2% | 11.0% | 46.2% | 11.2% | 11.2% | **53.5%** |
| | advCNN | 3.0% | 0.3% | 6.6% | 1.1% | 0.5% | **11.7%** | 10.5% | 2.9% | 10.5% | 2.9% | 2.9% | **42.9%** |
| | WideResNet | 1.3% | 1.1% | **1.7%** | 1.2% | 1.1% | 1.2% | 13.5% | 4.5% | 18.0% | 4.7% | 4.7% | **25.0%** |
| CIFAR-100 | vanCNN | 2.5% | 0.0% | **5.5%** | 0.3% | 0.0% | **5.5%** | 8.1% | 1.6% | 18.7% | 5.6% | 2.0% | **38.0%** |
| | advCNN | 1.1% | 0.0% | 1.1% | 0.4% | 0.4% | **1.2%** | 3.7% | 0.7% | 3.7% | 2.1% | 1.5% | **6.5%** |
| | WideResNet | 3.0% | 0.6% | 3.5% | 0.7% | 0.7% | **5.1%** | 17.0% | 9.7% | **19.9%** | 13.7% | 9.7% | **19.9%** |
| FMNIST | vanCNN | 0.5% | -0.0% | 0.5% | 0.3% | 0.3% | **0.8%** | 3.7% | 0.4% | 3.7% | 2.3% | 2.3% | **10.0%** |
| | advCNN | 1.0% | 0.1% | 1.3% | 1.0% | 0.7% | **6.4%** | 7.1% | 0.7% | 7.4% | 4.2% | 4.2% | **27.6%** |
| | WideResNet | 0.7% | 0.1% | 3.2% | 0.5% | 0.2% | **3.4%** | 5.6% | 1.1% | 5.9% | 1.2% | 1.2% | **28.0%** |
| STL-10 | vanCNN | 2.8% | 0.6% | 6.7% | 2.7% | 0.6% | **8.7%** | 16.4% | 3.2% | 18.5% | 11.6% | 3.2% | **34.1%** |
| | advCNN | 8.0% | 1.0% | 8.0% | 3.7% | 1.2% | **8.3%** | 16.8% | 6.1% | 16.8% | 9.8% | 7.3% | **35.4%** |
| | WideResNet | 2.3% | 0.4% | 3.6% | 2.2% | 0.4% | **3.9%** | 5.9% | 3.7% | 12.1% | 4.6% | 3.7% | **22.3%** |
| UTKFace | vanCNN | 0.5% | 0.1% | **1.1%** | 0.5% | 0.5% | **1.1%** | 4.0% | 1.4% | 8.1% | 4.0% | 2.2% | **9.6%** |
| | advCNN | 4.8% | 0.6% | 4.8% | 1.3% | 1.3% | **10.2%** | 12.9% | 4.5% | 12.9% | 6.7% | 5.8% | **43.1%** |
| | WideResNet | 0.5% | 0.1% | 0.7% | 0.2% | 0.2% | **1.1%** | 4.8% | 0.5% | 6.6% | 2.4% | 1.5% | **10.8%** |
| Adult | MLP | 0.5% | -0.0% | 0.5% | 0.4% | 0.4% | **0.9%** | 2.8% | -0.6% | 2.8% | 1.5% | 0.8% | **10.0%** |
| Location | MLP | 2.6% | 0.0% | 2.9% | 0.0% | 0.0% | **3.2%** | 9.5% | 1.3% | 10.4% | 1.4% | 1.3% | **10.8%** |
| Purchase | MLP | 2.0% | -0.1% | 2.1% | 1.3% | -0.1% | **2.7%** | 6.3% | 2.5% | 8.5% | 4.0% | 2.5% | **16.1%** |
| Texas | MLP | 0.3% | -0.1% | 1.5% | -0.1% | -0.1% | **2.7%** | 2.2% | 0.8% | 8.8% | 0.8% | 0.8% | **26.5%** |

## G Additional Results on Entropy / Prediction Uncertainty Distributions



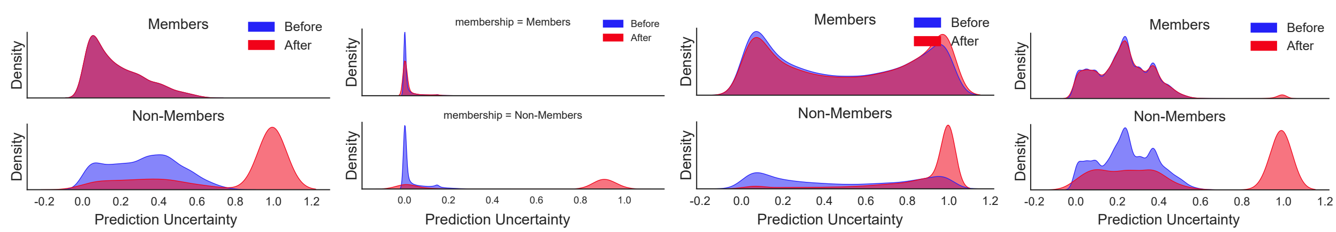

(a) CIFAR-100 (b) CIFAR-10 (c) FMNIST (d) STL-10

Fig. S5. Prediction uncertainty distributions (before/after apcMIA perturbation) for **AdvCNN**. Separation between members and non-members increases after perturbation.




(a) CIFAR-100 (b) CIFAR-10 (c) FMNIST (d) STL-10

Fig. S6. Prediction uncertainty distributions (before/after) for **VanCNN**.




(a) Location (b) Purchase (c) Adult (d) Texas

Fig. S7. Prediction uncertainty distributions for **MLP** on non-image datasets.

## H Robustness under DP–SGD

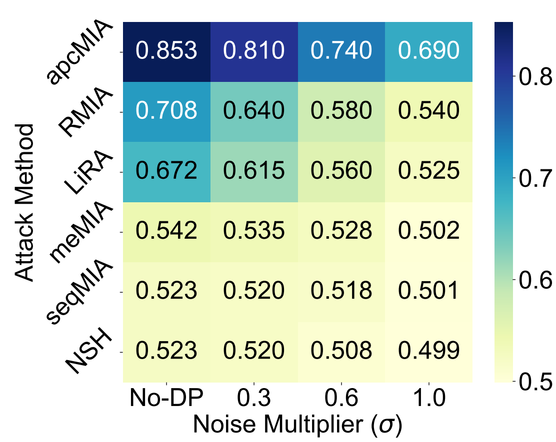



Fig. S8. Balanced accuracy of membership inference attacks against DP-trained target model on the Location dataset with varying noise multipliers ($\sigma$). `apcMIA` outperforms baselines across all privacy levels.

# I Gap-aware Sigmoid-based Schedule

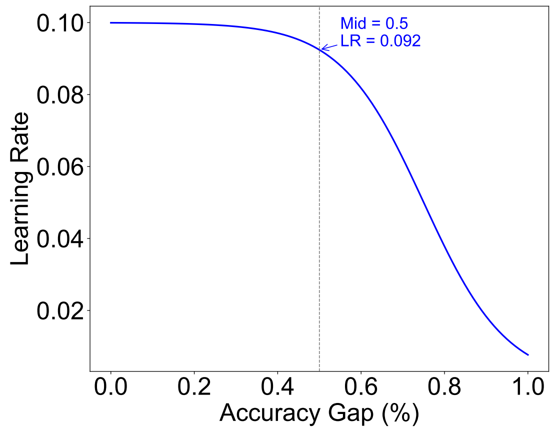
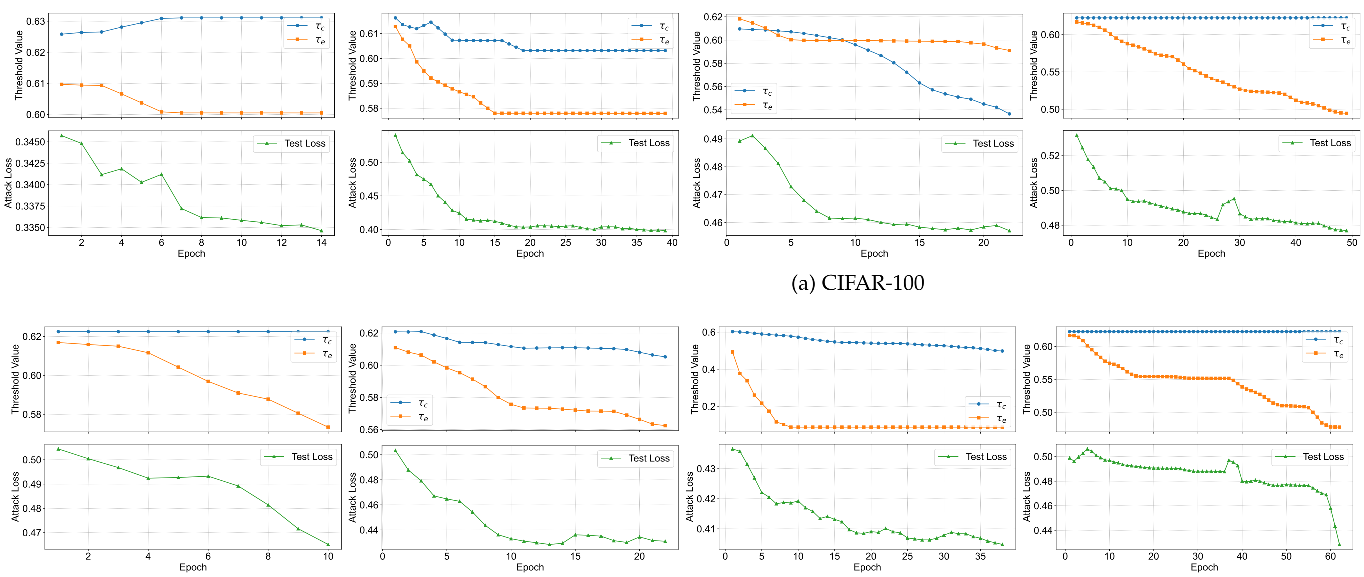



Fig. S9. Adaptive learning rate schedule based on an inverted-sigmoid function over the accuracy gap. Smaller gaps (well-generalized models) receive higher learning rates to accelerate threshold convergence, while larger gaps use lower rates for stable updates.

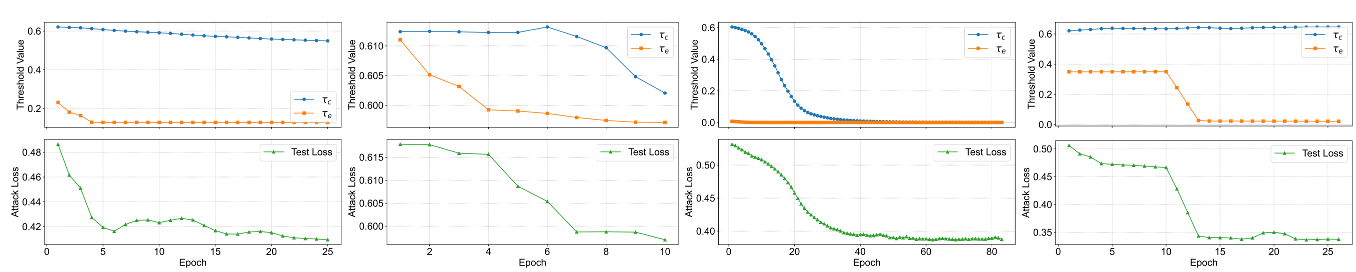
# J Evolution of Learned Thresholds




(a) CIFAR-100

(b) CIFAR-10 (c) STL-10 (d) CIFAR-100 (e) UTKFace

Fig. S10. Evolution of learned thresholds (cosine $\tau_c$, entropy $\tau_e$) and apcMIA test loss for image datasets under AdvCNN (top row) and WideResNet (bottom row).

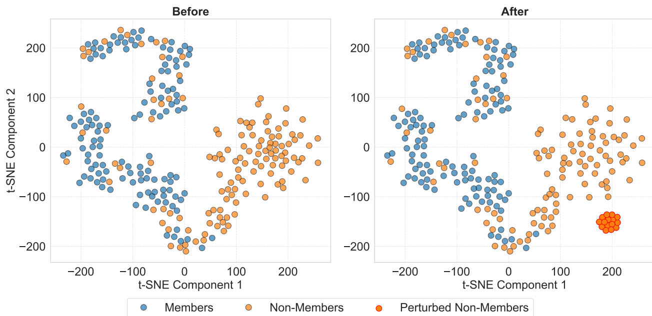



(a) Location (b) Purchase-100 (c) Texas-100 (d) UCI Adult

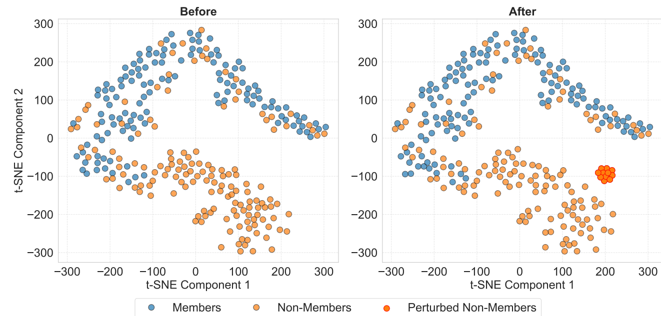
Fig. S11. Evolution of learned thresholds and apcMIA test loss for non-image datasets under MLP.

# K t-SNE Cluster Results

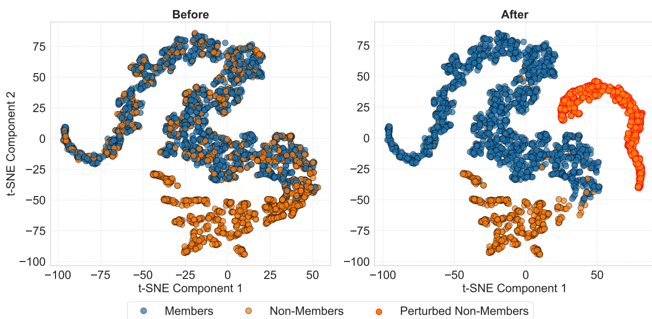
## K.1 Image Datasets



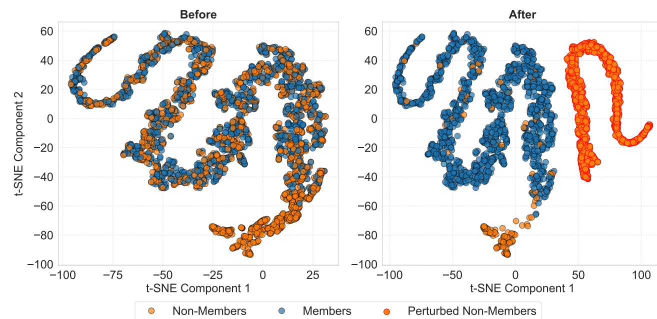

(a) Class 30



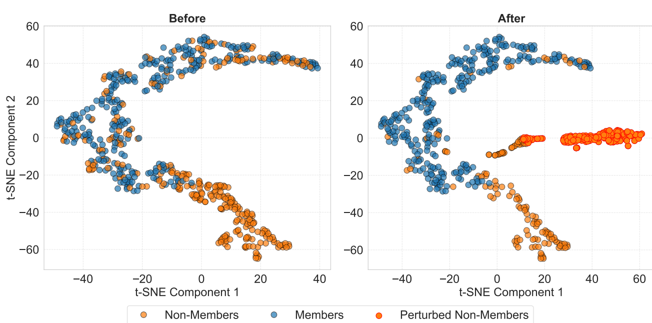

(b) Class 88



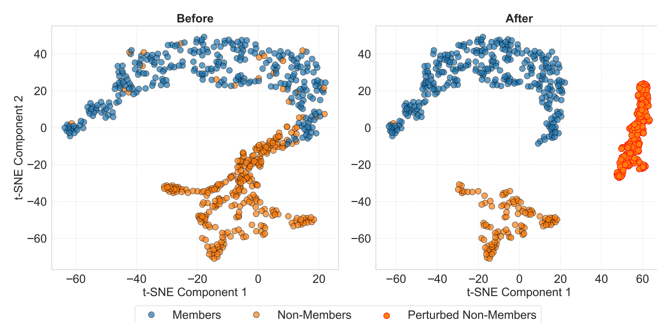

(c) Class 3



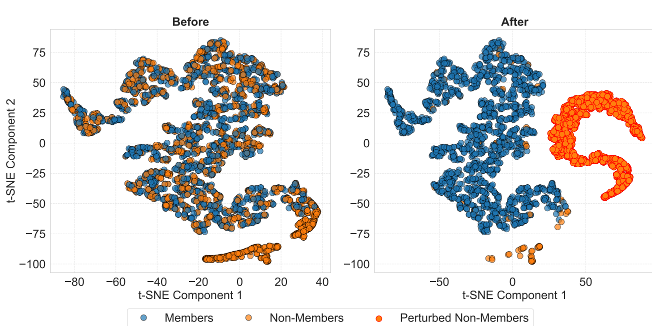

(d) Class 9



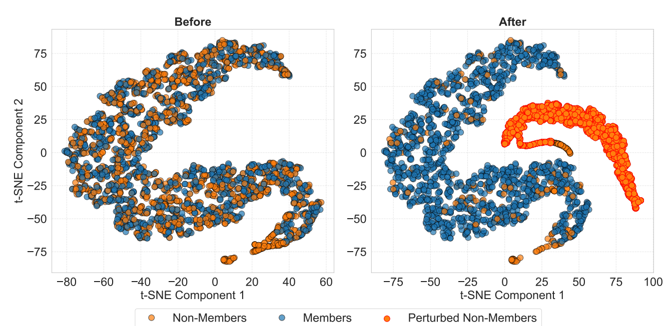

(e) Class 1




(f) Class 4



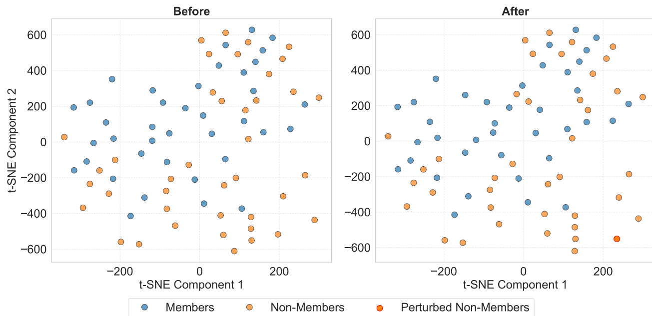

(g) Class 1



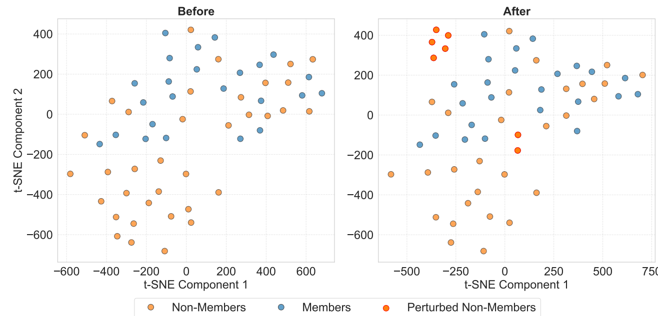

(h) Class 10

Fig. S12. t-SNE plots before/after `apcMIA` on AdvCNN models for CIFAR-100, CIFAR-10, STL-10, and FMNIST. Perturbed non-members (red) move away from members (blue), improving separability.

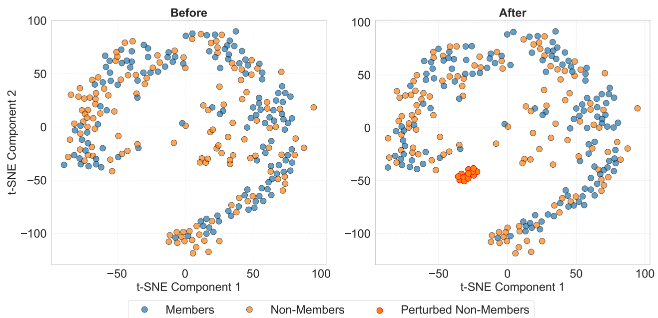
### K.2 Non-Image Datasets



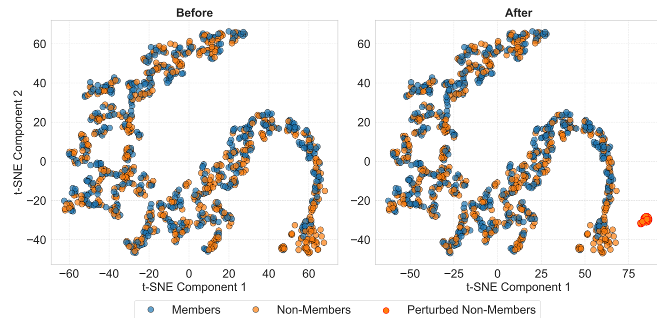

(a) Class 1 (Location)



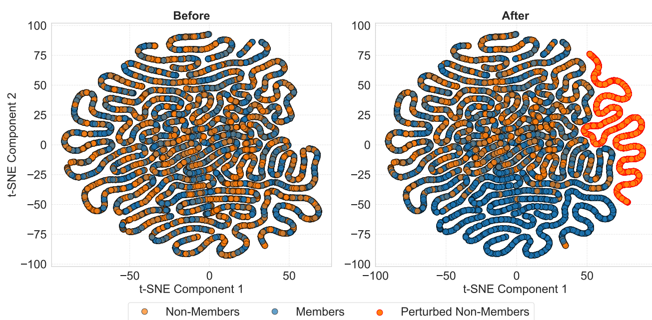

(b) Class 13 (Location)



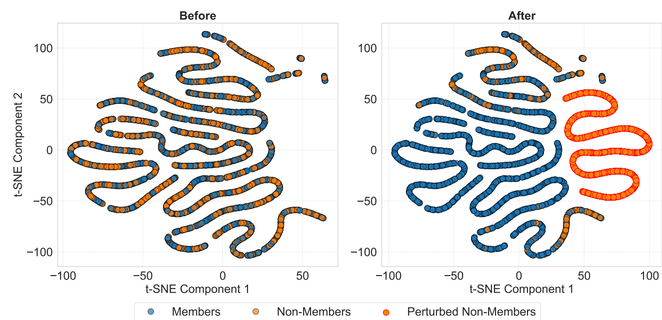

(c) Class 12 (Texas-100)



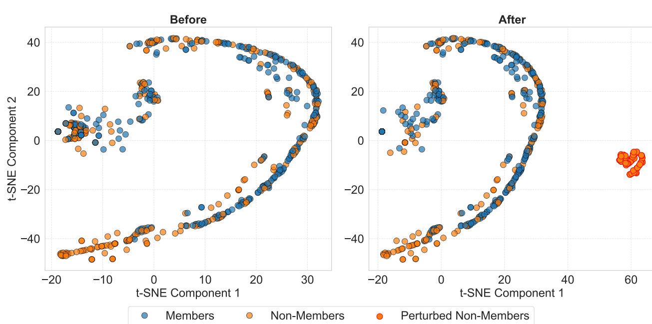

(d) Class 31 (Texas-100)



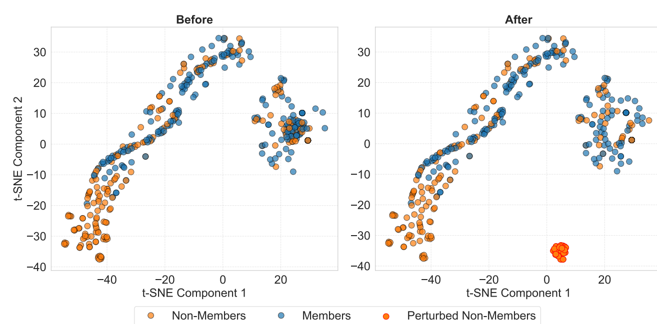

(e) Class 1 (Adult)




(f) Class 2 (Adult)




(g) Class 1 (Purchase-100)




(h) Class 97 (Purchase-100)

Fig. S13. t-SNE visualizations for non-image datasets under MLP. Left in each pair: before perturbation; right: after perturbing non-members.

## References

[1] S. Y. Zhang, Zhifei and H. Qi, "Age progression/regression by conditional adversarial autoencoder," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2017.

[2] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.

[3] A. Coates, A. Ng, and H. Lee, "An analysis of single-layer networks in unsupervised feature learning," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, G. Gordon, D. Dunson, and M. Dudík, Eds., vol. 15. Fort Lauderdale, FL, USA: PMLR, 11–13 Apr 2011, pp. 215–223. [Online]. Available: https://proceedings.mlr.press/v15/coates11a.html

[4] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms," 2017. [Online]. Available: https://arxiv.org/abs/1708.07747

[5] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership Inference Attacks Against Machine Learning Models," in *IEEE Symposium on Security and Privacy (S&P)*. IEEE, 2017, pp. 3–18.

[6] D. Yang, D. Zhang, and B. Qu, "Participatory cultural mapping based on collective behavior data in location-based social networks," *ACM Trans. Intell. Syst. Technol.*, vol. 7, no. 3, Jan. 2016. [Online]. Available: https://doi.org/10.1145/2814575

[7] Y. Liu, R. Wen, X. He, A. Salem, Z. Zhang, M. Backes, E. D. Cristofaro, M. Fritz, and Y. Zhang, "ML-Doctor: Holistic risk assessment of inference attacks against machine learning models," in *31st USENIX Security Symposium (USENIX Security 22)*. Boston, MA: USENIX Association, Aug. 2022, pp. 4525–4542. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/liu-yugeng

[8] Ander, "The best CNN for CIFAR10 from scratch (93% accuracy)," Feb. 2024. [Online]. Available: https://medium.com/@anderaquerretamontoro/the-best-cnn-for-cifar10-from-scratch-93-accuracy-bde35e17fca6

[9] S. Zarifzadeh, P. Liu, and R. Shokri, "Low-cost high-power membership inference attacks," 2024. [Online]. Available: https://arxiv.org/abs/2312.03262

[10] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership Inference Attacks From First Principles," *CoRR abs/2112.03570*, 2021.

[11] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy Risk in Machine Learning: Analyzing the Connection to Overfitting," in *IEEE Computer Security Foundations Symposium (CSF)*. IEEE, 2018, pp. 268–282.

[12] L. Melis, C. Song, E. D. Cristofaro, and V. Shmatikov, "Exploiting Unintended Feature Leakage in Collaborative Learning," in *IEEE Symposium on Security and Privacy (S&P)*. IEEE, 2019, pp. 497–512.

[13] N. Ullah, M. N. Aman, and B. Sikdar, "memia: Multilevel ensemble membership inference attack," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 1, pp. 93–106, 2025.

[14] A. Salem, Y. Zhang, M. Humbert, P. Berrang, M. Fritz, and M. Backes, "ML-Leaks: Model and Data Independent Membership Inference Attacks and Defenses on Machine Learning Models," in *Network and Distributed System Security Symposium (NDSS)*. Internet Society, 2019.

[15] Y. Liu and J. Xin, "Seqmia: Membership inference attacks against machine learning classifiers using sequential information," in *2022 4th International Conference on Applied Machine Learning (ICAML)*, 2022, pp. 31–35.

[16] L. Watson, C. Guo, G. Cormode, and A. Sablayrolles, "On the Importance of Difficulty Calibration in Membership Inference Attacks," *CoRR abs/2111.08440*, 2021.

[17] J. Ye, A. Maddi, S. K. Murakonda, and R. Shokri, "Enhanced Membership Inference Attacks against Machine Learning Models," *CoRR abs/2111.09679*, 2021.

[18] M. Nasr, R. Shokri, and A. Houmansadr, "Machine Learning with Membership Privacy using Adversarial Regularization," in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2018, pp. 634–646.